Synthetic Data for Named Entity Recognition in Sensitive Domains

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1 Introduction

The U.S. Navy's Composite Unit Training Exercise (COMPTUEX) is a three-week certification exercise performed by Carrier Strike Groups (CSG) before a combat deployment. The exercise is designed to assess unit readiness through increasingly complex scenarios requiring cooperation between surface, subsurface, and air components of the CSG. The exercise is led by an training group commander who is responsible for designing complex scenarios that often use a TREASURE COAST schema: an ontology of fictional nations, locations, and actors designed to emulate ally and adversary groups without using real-world actors. Training scenarios are written for every CSG for every predeployment phase, but are not made publically available due to their sensitive nature.

The purpose of this investigation is to determine whether a large language model can be used to develop a high-quality synthetic training dataset given minimal examples and whether the synthetic data can be used to train a Named Entity Recognition system.

2 Related Work

Previous work includes Evaluating Synthetic Data Generation from User Generated Text (Chim et al., 2025), which finds that language models are prone to generating sterilized versions of human text that don't accurately capture linguistic diversity in human generated content. Additionally, models trained on synthetic corpora tend to pick up on the generation logic patterns of the LLM and not the full semantic picture of the domain.

The conflibert model used in this experiment is detailed in (Hu et al., 2022). The authors remark that existing systems such as Integrated Crisis Early Warning System, used for conflict forecasting and research in political science, rely on dated pattern-matching techniques and are difficult to maintain. Hu et al. argue that an LLM designed for conflict and political violence was necessary to bridge the gap between older static methods of NER and newer, state of the art methods based on the transformer architecture.

3 Data

The synthetic dataset used in this experiment was generated by Microsoft Copilot using OpenAI GPT-5. Copilot was given one example document, an exercise announcement for Exercise Bold Alligator 2014 written using the TREASURE COAST schema, and asked to generate 1000 BIO-encoded example strings in CoNLL format. One of the first roadblocks in this experiment was data repetition; an examination of the generated examples revealed that CoPilot began to generate duplicate examples. Of the 550 examples generated, only 477 examples were unique. Data generation was cut off at this point.

Data was generated in chunks of twenty examples at a time to afford efficient auditing. Reviewing each chunk of examples led to the discover of patterns that were not ideal for the training data. Copilot was clearly sticking to a "Members of X did Y as part of operation Z" template. Numerous attempts were made to redirect prompt generation to cover different scenarios, vary the types of entities being generated, and create more complicated examples:

- Two O Pyropian B-NORP naval O mechanics O serviced O a O Hull-9 B-PLATFORM pressure I-PLATFORM sub I-PLATFORM in O the O Ashrift B-LOCATION Drydock I-LOCATION . O
- Captain B-PERSON Ravik I-PERSON oversaw O the O deployment O of O the O Amber-

forge B-ORG_UNIT Forward I-ORG_UNIT Detachment I-ORG_UNIT . O

Copilot was generally able to generate CoNLL-style annotated examples following the ontology provided. Additionally it appeared to be competent at mixing various entity types into examples. Auditing discovered a few cases of bad annotations, with I- labels following O- labels, and mismatched B- I- entities. Mention repair was handled in the evaluation process by seqeval. The generated data was split 70-15-15 into train, dev, and evaluation datasets.

In addition to the synthetic dataset, the Bold Alligator exercise announcement was also hand-annotated as a gold standard evaluation set. The models were each trained and evaluated on synthetic data, and then re-evaluated on the gold label dataset to assess the robustness of each model on real data.

4 Models

Three models were used in this experiment.

1. Distilbert-base-cased:link
2. Bert-base-cased:link
3. Conflibert-named-entity-recognition:link

Bert and Distilbert for Token Classification were chosen to assess ability of general token classification models used in NER tasks..These models are not trained in a specific domain but are trained to assign labels to tokens. Conflibert is a Bert model specifically finetuned on newswire conflict data and was used to assess whether a domain-specific model performed better on the task.

Each model was finetuned on the train split of the synthetic dataset over 30 epochs, evaluated on the test split of the synthetic dataset, and finally evaluated on the gold label dataset. Conflibert showed the highest performance on the gold label data and was further tuned by a grid search over epochs, batch size, learning rate, and weight decay.

5 Results

Results were a mixture of expected and surprising in the initial best model assessment. As expected, Conflibert scored the highest F1 on both the synthetic and gold datasets. On the gold label set, however, distilbert scored a significantly higher F1 than Bert-base-cased.

5.1 Best Model Selection

distilbert	synthetic	gold
F1	96.67	41.82
Accuracy	99.36	90.14

bert-based-cased	synthetic	gold
F1	98.07	27.27
Accuracy	99.79	86.78

conflibert	synthetic	gold
F1	98.34	46.85
Accuracy	99.36	87.74

Accuracy is reported in these tables because the gold label accuracy scores were suprisingly low compared to performance on synthetic data.

5.2 Grid Search on Best Model

The criteria decided for best model was the model that received the highest F1 score on the gold label data. Conflibert unsurprisingly outperformed the other two models.

Epochs	10
Batch Size	16
Learning Rate	5e-5
Weight Decay	0.01
F1	**56.07**

Table 1: Grid Search Best Hyperparameter Results on Conflibert

Grid searching resulted in a significant performance boost for Conflibert, gaining nearly ten points on F1.

6 Discussion

The results of this experiment indicate that small synthetic datasets are not sufficient to train a model that is capable of performing NER tasks on real-world data. All three models in this experiment performed well on the synthetic dataset, but performance dropped significantly on the gold label data which was chosen specifically as human generated data that a model would see.

The synthetic data generated by the LLm was monotonous and tended to follow a very specific template, despite repeated attempts to redirect the output during generation.

7 Conclusion

There are many separate approaches to strengthening the results of this project, but many of them are intertwined in some way that makes piecewise improvement difficult. An intuitive first approach would be to vastly increase the size of the training dataset. Doubling or even tripling the number of training examples would be ideal but this raises other issues.

The models' high performance on the training data suggests that they don't actually need more examples. Accuracy and F1 scores in the high-90s indicate that the data was essentially being memorized by the model. The sole benefit to adding more training would be to determine whether more data leads to performance improvement on real world data. This leads to the conclusion that massive amounts of data need to be generated to see any improvement and suggests that synthetic datasets are not an *efficient* way to perform NER on scarce data.

One of the struggles during the data generation phase of this experiment was the repetitiveness of data. The generated data would benefit greatly from a variety of structures and cadences that more faithfully represent real data. This requires the generator LLM to create and then annotate documents in a variety of formats such as news articles, secure messages and emails, after-action reports, or mission briefings. The synthetic data somewhat effectively represented the formulaic appearance of real military documents and bulletins, but those documents also tend to feature heavy use of acronyms in uppercase text which further complicates the NER task.

Conflibert's superior performance over general domain models and the massive performance boost from hyperparameter tuning confirm that finetuned models will outperform out-of-the-box models in niche tasks. This is not a surprising discovery. Finetuning bert-base-cased on a large set existing documents, and further training on scenario-specific literature would no doubt achieve desired results, but the purpose of this experiment was to ascertain whether this task could be achieved despite data scarcity.

A LLM Prompting

- Model: Microsoft Copilot with OpenAI GPT-5
- Prompt: Generate 1000 examples for an NER task using the provided Bold-Alligator document as a guideline for format. Follow the TREASURE COAST scenario schema for entities. Use the following formatting guidelines:

  label schema: O

  B-PERSON, I-PERSON

  B-ORG_UNIT, I-ORG_UNIT

  B-NORP, I-NORP

  B-LOCATION, I-LOCATION

  B-PLATFORM, I-PLATFORM

  B-EVENT, I-EVENT

  output format: CoNll Format

  split: train only

B Ontology

- Events: Used for operation names such as "Operation Glassward".
- Locations: Any geographical location.
- NORP: Nationality designations such as Garnetian.
- Organizations: Refers to named units such as 5th Mountain Brigade, CSG-2.
- People: Mentions of individual people to include rank or title, and name.
- Platforms: Includes any weapon system or named vehicle or equipment, such as USS Arleigh Burke or DDG-51.

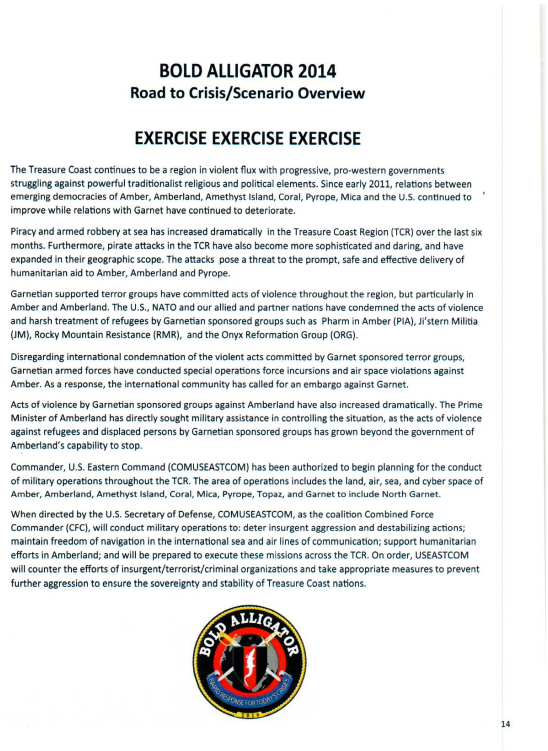
C Encoding Schema

Entity	Begin	Intermediate
EVENT	B-EVENT	I-EVENT
LOCATION	B-LOCATION	I-LOCATION
NORP	B-NORP	I-NORP
ORGANIZATIONS	B-ORG-UNIT	I-ORG-UNIT
PEOPLE	B-PERSON	I-PERSON
PLATFORMS	B-PLATFORM	I-PLATFORM

D Grid Search Hyperparameters

- Batch Size: 16,32,64
- Epochs: 5,10,15,20,25,30
- Learning Rate: 5e-5,4e-5,3e-5,2e-5,1e-5
- Weight Decay: 0.0,0.01,0.1

E Exercise Bold Alligator 2014



BOLD ALLIGATOR 2014
Road to Crisis/Scenario Overview

EXERCISE EXERCISE EXERCISE

The Treasure Coast continues to be a region in violent flux with progressive, pro-western governments struggling against powerful traditionalist religious and political elements. Since early 2011, relations between emerging democracies of Amber, Amberland, Amethyst Island, Coral, Pyrope, Mica and the U.S. continued to improve while relations with Garnet have continued to deteriorate.

Piracy and armed robbery at sea has increased dramatically in the Treasure Coast Region (TCR) over the last six months. Furthermore, pirate attacks in the TCR have also become more sophisticated and daring, and have expanded in their geographic scope. The attacks pose a threat to the prompt, safe and effective delivery of humanitarian aid to Amber, Amberland and Pyrope.

Garnetian supported terror groups have committed acts of violence throughout the region, but particularly in Amber and Amberland. The U.S., NATO and our allied and partner nations have condemned the acts of violence and harsh treatment of refugees by Garnetian sponsored groups such as Pharm in Amber (PIA), Ji'stern Militia (JM), Rocky Mountain Resistance (RMR), and the Onyx Reformation Group (ORG).

Disregarding international condemnation of the violent acts committed by Garnet sponsored terror groups, Garnetian armed forces have conducted special operations force incursions and air space violations against Amber. As a response, the international community has called for an embargo against Garnet.

Acts of violence by Garnetian sponsored groups against Amberland have also increased dramatically. The Prime Minister of Amberland has directly sought military assistance in controlling the situation, as the acts of violence against refugees and displaced persons by Garnetian sponsored groups has grown beyond the government of Amberland's capability to stop.

Commander, U.S. Eastern Command (COMUSEASTCOM) has been authorized to begin planning for the conduct of military operations throughout the TCR. The area of operations includes the land, air, sea, and cyber space of Amber, Amberland, Amethyst Island, Coral, Mica, Pyrope, Topaz, and Garnet to include North Garnet.

When directed by the U.S. Secretary of Defense, COMUSEASTCOM, as the coalition Combined Force Commander (CFC), will conduct military operations to: deter insurgent aggression and destabilizing actions; maintain freedom of navigation in the international sea and air lines of communication; support humanitarian efforts in Amberland; and will be prepared to execute these missions across the TCR. On order, USEASTCOM will counter the efforts of insurgent/terrorist/criminal organizations and take appropriate measures to prevent further aggression to ensure the sovereignty and stability of Treasure Coast nations.

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Figure 1: Gold label document for the Bold Alligator 2014 exercise.

References

Jenny Chim, Julia Ive, and Maria Liakata. 2025. Evaluating synthetic data generation from user generated text. *Computational Linguistics*, 51(1):191–233.

Yifei Hu, Zining Li, Vid Vukovic, and Andrew Halterman. 2022. ConfliBERT: A pre-trained language model for political conflict and violence. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 5469–5482, Seattle, United States. Association for Computational Linguistics.